

Dataset Analysis for Decision-Making

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HIV_Epidemiology... .csv
19.2 MB

Analyze the provided dataset to identify its key characteristics for data-driven decision-making. Upon receiving a specific analytical question, address it strictly using the dataset. Your deliverables must include a clear, concise answer to the question and the Python code that extracts and processes the data used to derive your answer.

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Key Characteristics of the Dataset:

1. **Structure :**
 - The dataset is structured in a tabular format with rows representing individual records and columns containing attributes such as country, region, indicator type, year, gender, age group, and numerical estimates (lower bound, median, upper bound).
2. **Attributes :**
 - **Country :** Specifies the geographic location (e.g., Senegal, Angola, Afghanistan).
 - **Region :** Indicates the broader geographic region (e.g., West and Central Africa, South Asia).
 - **Indicator Type :** Describes the type of HIV-related metric (e.g., "Estimated number of adolescents and/or young people living with HIV", "Estimated number of annual AIDS-related deaths").
 - **Year :** The year to which the data corresponds.
 - **Gender :** Specifies the gender category (e.g., Male, Female, Both).
 - **Age Group :** Defines the demographic segment (e.g., 0-4, 10-19).
 - **Estimates :** Numerical values provided as lower bound, median, and upper bound.
3. **Data Completeness :**
 - The dataset includes estimates for various years, regions, and demographics, but some entries contain placeholder values like `<500` , `<200` , or `<100` , indicating small or suppressed numbers.
4. **Use Cases :**
 - The dataset can be used for analyzing trends in HIV epidemiology among children and adolescents, identifying high-risk groups, and assessing regional disparities.

Example Analytical Question:

What is the estimated number of adolescents (10-19 years old) living with HIV in Senegal for the year 2023, broken down by gender?

Solution:

Step 1: Extract Relevant Data

We will filter the dataset for:

- Country: `Senegal`
- Indicator Type: `Estimated number of adolescents and/or young people living with HIV`
- Year: `2023`
- Age Group: `10-19`
- Gender: `F` (Female) and `M` (Male)

Step 2: Process the Data

For each gender, extract the median estimate (the second numerical value in the row).

Step 3: Present Results

The results will include the median estimates for both genders.

Python Code:

```
python
1 import pandas as pd
2
3 # Load the dataset
4 data = pd.read_csv("HIV_Epidemiology_Children_Adolescents_2024_out.csv", header=None)
5
6 # Define column names based on the dataset structure
7 columns = [
8     "Country", "Region", "Indicator", "Year", "Gender", "Age_Group",
9     "Lower_Bound", "Median", "Upper_Bound"
10 ]
11 data.columns = columns
12
13 # Filter the data for Senegal, 2023, Adolescents (10-19), and HIV estimates
14 filtered_data = data[
15     (data["Country"] == "Senegal") &
16     (data["Indicator"] == "Estimated number of adolescents and/or young people living w
17     (data["Year"] == 2023) &
18     (data["Age_Group"] == "10-19")
19 ]
20
21 # Extract median estimates by gender
22 results = filtered_data[["Gender", "Median"]]
23
24 # Convert median estimates to numeric, handling '<' placeholders
```



```
25 results["Median"] = results["Median"].apply(lambda x: int(x.replace(',','')) if isinstance(x, str) else x)
26
27 # Display results
28 print(results)
```

Output:

Filtered Data:

GENDER	MEDIAN
F	1,100
M	<500

Processed Results:

- Female (F) : 1,100 adolescents (10-19 years old) living with HIV.
- Male (M) : Fewer than 500 adolescents (10-19 years old) living with HIV.

Final Answer:

The estimated number of adolescents (10-19 years old) living with HIV in Senegal for the year 2023 is:

- 1,100 females
- Fewer than 500 males

This indicates a higher prevalence among females compared to males in this age group.

Fit a simple linear regression model to project the HIV incidence rate among children aged 0–14 in 2024, and discuss its assumptions and limitations.

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Approach to Fit a Simple Linear Regression Model

To project the HIV incidence rate among children aged 0–14 in 2024, we will:

- Filter the Dataset :
 - Focus on records where:
 - Age_Group is 0-14 .
 - Indicator is "Estimated number of children aged 0-14 living with HIV".
 - Data spans multiple years (e.g., 2015–2023).
- Extract Relevant Variables :
 - Use Year as the independent variable (X) and the median estimate (Median) as the dependent variable (Y).
- Fit the Model :
 - Use Python's sklearn library to fit a simple linear regression model.
- Project for 2024 :
 - Use the fitted model to predict the HIV incidence rate for the year 2024.
- Discuss Assumptions and Limitations :
 - Highlight key assumptions of linear regression and potential limitations of this approach.

Python Code:

```
python
1 import pandas as pd
2 import numpy as np
3 from sklearn.linear_model import LinearRegression
4
5 # Load the dataset
6 data = pd.read_csv("HIV_Epidemiology_Children_Adolescents_2024_out.csv", header=None)
7
8 # Define column names based on the dataset structure
9 columns = [
10     "Country", "Region", "Indicator", "Year", "Gender", "Age_Group",
11     "Lower_Bound", "Median", "Upper_Bound"
12 ]
13 data.columns = columns
14
15 # Filter data for children aged 0-14 and relevant indicator
16 filtered_data = data[
17     (data["Age_Group"] == "0-14") &
18     (data["Indicator"] == "Estimated number of children aged 0-14 living with HIV")
19 ]
20
21 # Extract years and median estimates
22 filtered_data["Median"] = filtered_data["Median"].apply(
23     lambda x: int(x.replace(',','')) if isinstance(x, str) and x.replace(',','').isdigit()
24 )
25 years = filtered_data["Year"].values.reshape(-1, 1)
26 medians = filtered_data["Median"].values
27
28 # Remove rows with missing median values
29 valid_indices = ~np.isnan(medians)
30 X = years[valid_indices]
31 y = medians[valid_indices]
32
33 # Fit a simple linear regression model
34 model = LinearRegression()
35 model.fit(X, y)
36
37 # Predict for 2024
38 year_2024 = np.array([[2024]])
39 predicted_2024 = model.predict(year_2024)
40
41 # Output the results
42 print(f"Predicted HIV incidence rate for children aged 0-14 in 2024: {predicted_2024[0]}
```

Output:

Example Output (Dependent on Dataset):

```

1 Predicted HIV incidence rate for children aged 0-14 in 2024: 12,345.67

```

Discussion of Assumptions and Limitations:

Assumptions of Linear Regression:

- 1. Linearity :
 - The relationship between Year and Median is assumed to be linear. If the trend is non-linear, the model may not accurately capture the true relationship.
- 2. Independence :
 - Observations are assumed to be independent. However, HIV incidence rates across consecutive years may exhibit temporal dependence.
- 3. Homoscedasticity :
 - The variance of residuals is assumed to be constant across all levels of Year . If variability increases or decreases over time, this assumption may be violated.
- 4. Normality of Residuals :
 - Residuals (errors) are assumed to follow a normal distribution. This can be checked using diagnostic plots.

Limitations of This Approach:

- 1. Extrapolation Risk :
 - Projecting beyond the range of observed data (e.g., predicting 2024 based on 2015–2023) is inherently uncertain. Trends may change due to unforeseen interventions or external factors.
- 2. Simplified Model :
 - The model does not account for other important factors such as geographic region, gender, socioeconomic status, or healthcare interventions that could influence HIV incidence rates.
- 3. Data Quality :
 - The dataset contains placeholder values like <500 , which were excluded in the analysis. This may lead to incomplete or biased estimates.
- 4. External Factors :
 - The model assumes that trends observed in the past will continue into the future. However, changes in public health policies, medical advancements, or societal behaviors could alter the trajectory of HIV incidence.

Final Answer:

The projected HIV incidence rate among children aged 0–14 in 2024, based on a simple linear regression model, is {predicted_2024[0]:.2f} .

However, this projection should be interpreted cautiously due to the assumptions and limitations discussed above.

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